

3 Energy



LEARNING OUTCOMES

Students will be able to:

- ▶ define energy.
- ▶ express energy in proper units.
- ▶ discuss about different forms of energy.
- ▶ define conversion of energy from one form to another in different situations.
- ▶ state the law of conservation of energy, with examples.

Energy

Energy is a commonly-used term in our everyday life. We talk of energy in terms of physical energy of an individual, by the amount of work a body can do.

Thus, energy is the capacity to do work. This energy can be due to its state of motion, position or its configuration. An object, when raised to a certain height or made to move, possesses a certain amount of energy. Energy and work, being of the same nature, are expressed by the same units.

Units of Energy

SI unit of energy is joule (J).

CGS unit of energy is erg.

$$1 \text{ J} = 10^7 \text{ erg}$$

Heat energy is usually measured in *calorie*.
 $1 \text{ calorie} = 4.2 \text{ J}$

1 calorie is the energy (heat) required to raise the temperature of 1 g of water by 1°C .

Other units are watt hour, kilowatt hour and electron volt.

Forms of Energy

Energy appears in many forms. It can be converted from one form to another, but it cannot be created or destroyed. The different forms of energy are discussed below.

1. Mechanical Energy

It is the energy possessed by a body due to its position, configuration or state of motion.

It can be divided into two main categories:

- i. Kinetic energy
- ii. Potential energy

The total mechanical energy of a body is the sum of its potential and kinetic energy.

i. Kinetic Energy

A moving hammer performs work when it strikes a nail. Similarly, when a bullet is fired, it penetrates a thick wall. In each case, the object in motion has the ability to do work and thus possesses some amount of energy. This energy is called kinetic energy of the body. Hence, kinetic energy of a body depends on its mass and the speed with which it moves. A big stone with less speed can break a glass. Similarly, a small bullet with a high speed can cause a similar damage. In the first case, the stone has a greater mass, while in the second case, the bullet has less mass but greater speed. A ball, when kicked, is set in motion. It possesses energy due to its state of motion. Similarly, blowing wind has the energy to move a tree or a roof of a house away. A windmill functions on the same principle.

Therefore, kinetic energy of a body can be defined as the energy that a body possesses by virtue of its motion.

EXTRA FACT

The word 'kinetic' comes from the Greek word *kinesis*, which means motion.



(a) Windmill



(b) Football player



(c) A blowing wind

Fig. 3.1 Examples of bodies possessing kinetic energy

Examples of objects having kinetic energy are a moving bullet, a running child, a rolling stone, a moving car, flowing water, a running train, etc.

Kinetic energy can be calculated by the following formula:

$$KE = \frac{1}{2} \times \text{mass} \times (\text{velocity})^2$$

or, $KE = \frac{1}{2} mv^2$

If the mass of an object is doubled, the kinetic energy doubles. If the speed of an object is doubled, the kinetic energy increases by four times.

EXPERIMENT-1

A moving body possesses kinetic energy

Take a piece of cardboard. Bend it to give a V shape and place it on a plane surface as shown in the figure. At the base of the inclined plane, place a matchbox.

- Now, take a small sized marble and place it at position B on the inclined plane and release it. The marble rolls down the inclined plane and hits the matchbox. Measure the distance 'd' through which the matchbox moves.
- Repeat the experiment by placing the marble at position A on the inclined plane and measure the distance moved by the matchbox. You will find that in the second case the matchbox moves a greater distance.
- Take a bigger marble and place it at A on the inclined plane and release it. It will roll down and hit the matchbox. Measure the distance moved by the matchbox. You will find that the distance covered by the matchbox in the third case is more than that in the second case.

The experiment proves that—

- A body in motion possesses kinetic energy.
- Greater the speed of the body, greater is the kinetic energy.
- The kinetic energy of the body increases as its mass increases.

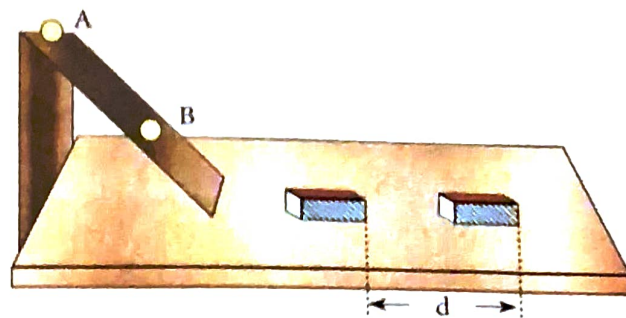


Fig. 3.2 A moving marble has kinetic energy

EXPERIMENT-2

Kinetic energy increases with increase in speed of the body

Place a wooden block on a table. Take a ball and keep it on the table, about 30 cm away from the block. Now, push the ball gently towards the block. The ball will collide with the block and the block will be displaced. Measure the displacement. Now, place the block and the ball in their previous positions again and push the ball harder so that it hits the block with a greater speed. You will find that the ball now pushes the block forward by a greater distance. Measure the displacement.

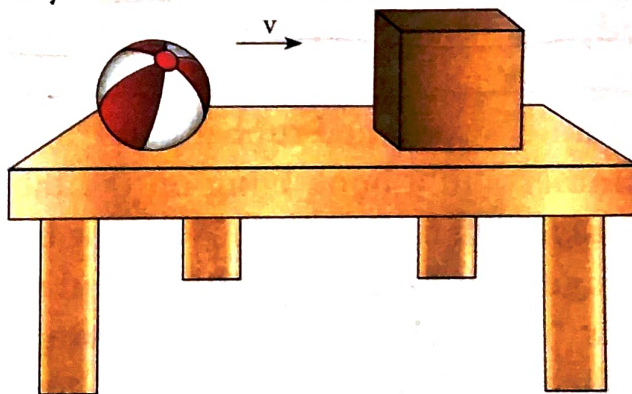


Fig. 3.3 Kinetic energy of the ball displaces the block

We conclude that greater the speed of the ball, more is the capacity to do work, i.e., a body has more kinetic energy when its speed is greater.

EXPERIMENT-3

Kinetic energy increases with increase in the mass of the body

Place a nail on a wooden block and hit it hard with a light hammer. The hammer has kinetic energy because of its motion. Thus, work will be done by the hammer on the nail. It will move the nail through some depth within the wooden block.

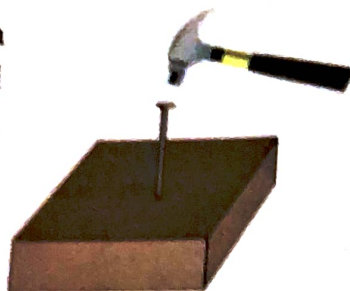


Fig. 3.4 Kinetic energy of the hammer drives a nail into the wooden block

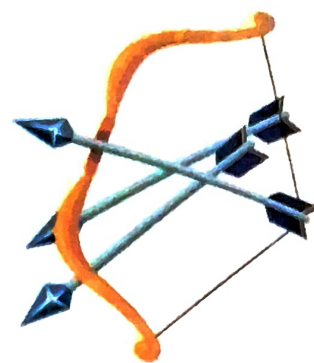
Now, take a heavier hammer and again place another identical nail and hit it with the same speed. You will observe that in the second case the nail will sink more into the wooden block.

This proves that greater the mass of a body, greater is the kinetic energy.

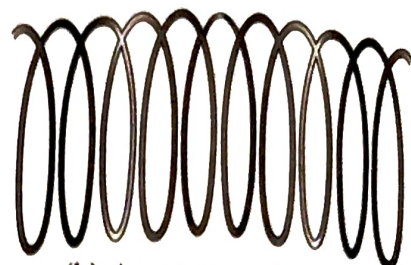
ii. Potential Energy

Potential energy is the energy stored in a body or a system as a result of its position, shape or change in configuration.

For example, a stretched rubber band, a raised hammer and a book lying on the table—all have potential energy. All these bodies change their form (or position) and regain their original form on removing the applied force. They have energy stored in them that helps them regain their original form. Here are some objects which possess potential energy due to their position and form (shape and size).



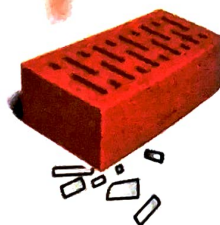
(a) A stretched bow



(b) A compressed spring



(c) A stretched elastic band of a catapult



(d) A brick raised to a certain height has the potential energy to break the chalks into pieces

Fig. 3.5 Some examples demonstrating potential energy

EXPERIMENT-4

Potential energy possessed by a body due to its shape and size (form)

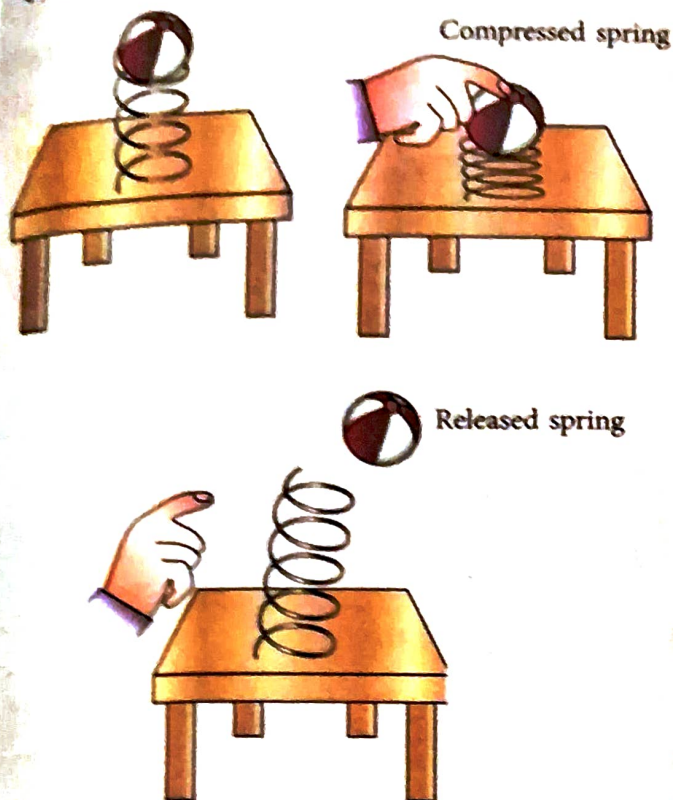


Fig. 3.6 Potential energy in a compressed spring

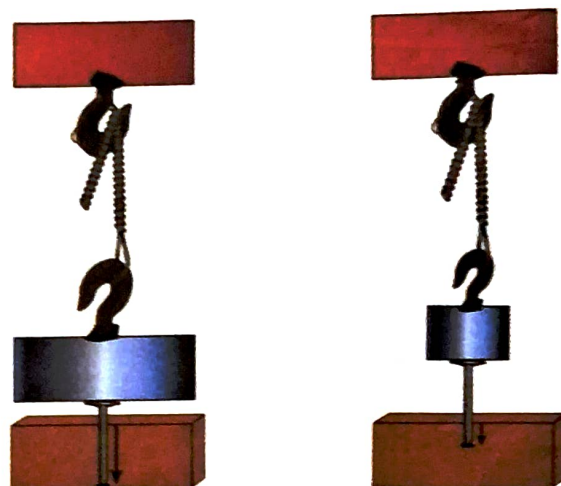
When you take an ordinary coiled spring and place a ball on it, it remains there. There is no change in its state or form. But when the ball is placed on the compressed spring and the spring is released, the ball bounces away. Why does it happen? When the spring is compressed, it gains energy which is stored in it. It is the stored energy due to which it bounces the ball away from it. Hence, the coiled spring possesses energy due to its form (shape and size). This energy is called its potential energy.

Measuring Potential Energy of a Body

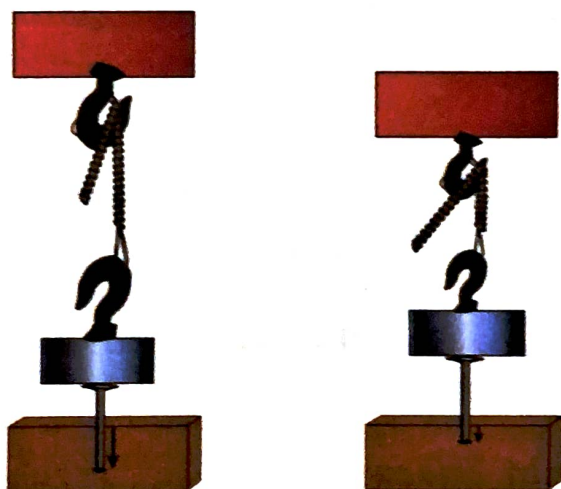
In fig. 3.7 (a), two objects of different masses when dropped from the same height make different impacts. The object

with greater mass drives the nail deeper into the wooden box than the object with less mass. Similarly, in fig. 3.7 (b), when two objects of similar masses are dropped from different heights, the object dropped from the greater height drives the nail deeper into the wooden box. Why does this happen? It is because of the difference in potential energy in both the cases.

Thus, we can conclude that potential energy depends upon the mass of the object and the height to which it is raised.



(a) Greater the mass, greater is the work done



(b) Greater the height, greater is the work done

Fig. 3.7 Potential energy depends upon mass and height

If 'm' is the mass of the body and 'h' is the height to which it is raised, then the potential energy of the body (PE) will be given by—

$$PE = mgh$$

Where 'g' is the acceleration due to gravity.

Potential energy is of two types — gravitational potential energy and elastic potential energy.

Gravitational potential energy is the energy possessed by an object when it is lifted above the surface of the earth. It increases along with increase in height of the object. For example, an apple hanging from a tree has gravitational potential energy due to its height.

Elastic potential energy is the energy possessed by an object when it is compressed, stretched or bent. For example, a stretched rubber band has elastic potential energy.

2. Heat Energy

Heat is a form of energy. We obtain heat energy by burning coal, wood, oil or liquefied petroleum gas (LPG). The energy obtained from them is utilised for various purposes. When water boils, it changes into steam. Steam is used in steam engines. Petrol or diesel on burning releases energy that runs a vehicle. Nowadays, we use compressed natural gas (CNG) for running our vehicles.



Fig. 3.8 Burning wood provides heat energy

3. Light Energy

Light is also a form of energy. Light energy is used by green plants for the process of

photosynthesis (formation of food). Light energy enables us to see the objects around us. It is also required for recording of images in a photographic film. Light energy is used in solar panels to derive energy of the sun to perform domestic functions.

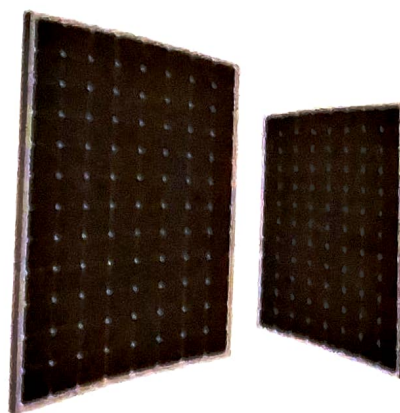


Fig. 3.9 Solar panels derive the energy of the sun

4. Electrical Energy

Electrical energy is one of the most useful forms of energy. It is used to produce heat (in a heater) and light (in a bulb and tube). It is also used to produce motion (such as to rotate fans, to run trains, trams, etc.). Electrical energy makes an electric bell produce sound.



(a) Electric bulb



(b) Electric heater

Fig. 3.10 Electric bulb and electric heater use electrical energy to glow

5. Magnetic Energy

The energy possessed by a magnet is called magnetic energy. A magnet can attract iron filings or pins towards it and hence

can produce motion; therefore it has the ability to do work. Most of the electrical appliances used at home like electric fans, doorbells, microwave ovens, mixers and grinders, televisions, telephones, mobile phones, etc. use magnetic energy for their working.



Fig. 3.11 Magnets are used in communication devices

6. Chemical Energy

Chemical energy is the energy stored in an object or material and is released during chemical reactions. Energy stored in food is converted into muscular energy and helps us to work. Chemical energy stored in petrol and diesel helps a car to run. Chemical energy stored in an electric cell is used to light a bulb or to run a fan.



Fig. 3.12 A battery has chemical energy stored in it

7. Sound Energy

It is also a form of energy. It is produced by the vibration of a body. When sound waves strike the ear membrane, it begins to vibrate and produces a sensation of hearing. When we pluck the string of a sitar or a violin, sound is produced. Similarly, when there is a loud clap of thunder or the blast of crackers

during Diwali, the doors and windows of the house rattle. Loudspeaker also produces sound when electric current is passed through it.



Fig. 3.13 Loudspeaker produces sound

8. Nuclear Energy

An atom of an element is mainly made of electrons, protons and neutrons. Protons and neutrons are located at the centre of the atom which is known as the nucleus. Electrons move around the nucleus in a spiral path called orbit.

The nucleus of an atom has an enormous amount of energy, which holds the protons and neutrons together. In an atomic power plant, when the nucleus of an element like uranium is split apart, tremendous amount of energy is released in the form of heat. This energy is called nuclear energy. Nuclear energy can be used for constructive as well as destructive purposes. In an atom bomb, nuclear energy is released which can be very destructive in nature. It is also used in the treatment of diseases like cancer.

**James Prescott
Joule**
(1818 – 1889)



Joule was a British physicist. He discovered the Law of Conservation of Energy with two other German physicists. In 1840, he derived a law, called Joule's Law. The international unit of energy, joule, is named in his honour.

KNOW YOUR SCIENTIST

Law of Conservation of Energy

The principle of conservation of energy states that energy can neither be created nor destroyed. The total energy of the universe remains constant. Energy can only be transformed from one form to another.

When an apple falls from a height, its gravitational potential energy decreases. In fact, the moment it is dropped, its potential energy is converted into kinetic energy. So, one form of energy is transformed into another. As a result, the total energy remains constant.

Transformation of Energy

Energy can be converted from one form to another. In each case, the transformation of energy is accompanied by the performance of work. Let us take a few examples:

i. Electrical energy into Sound energy

Loudspeaker, electric bell and stereo system are some of the many electrical appliances that convert electrical energy into sound energy.

ii. Heat energy into Electrical energy

Thermocouple, thermoelectric power stations, etc. convert heat energy into electrical energy.

iii. Solar energy into Heat energy

Solar cooker and solar furnaces convert solar energy into heat energy.

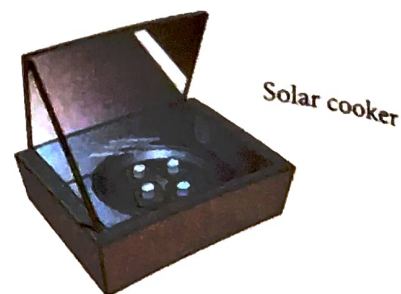


Fig. 3.14 Solar energy $\rightarrow$ Heat energy

iv. Electrical energy into Heat energy

Electric heater, electric iron, room heater, geyser, toaster, etc. convert electric energy into heat energy.



Electric kettle

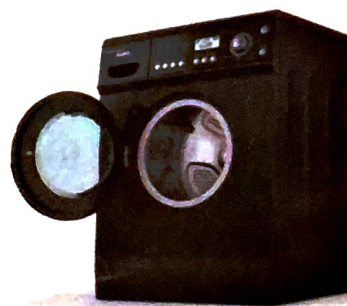


Room heater

Fig. 3.15 Electric energy $\rightarrow$ Heat energy

v. Electrical energy into Mechanical energy

Table fan, washing machine and other electric motors convert electrical energy into mechanical energy.



(a) Washing machine



(b) Table fan

Fig. 3.16 Electrical energy $\rightarrow$ Mechanical energy

vi. Electrical energy into Light energy

Tubes, bulbs, etc. convert electrical energy into light energy.



Electric bulb

Fig. 3.17 Electrical energy $\rightarrow$ Light energy

EXTRA FACT

Conventional light bulbs produce light using 10 per cent of the energy supplied. The rest of the energy is converted to heat. On the other hand, compact fluorescent light bulbs use about 80 per cent less electricity than conventional bulbs and last 12 times longer.

Light energy into Electrical energy

Photocell is an example that converts light energy into electrical energy.

Electrical energy into Chemical energy

While charging a battery, electrical energy is converted into chemical energy.

Chemical energy into Electrical energy

In a dry cell, chemical energy is converted into electrical energy. The same process also occurs in car batteries.

Mechanical energy into Electrical energy

A dynamo is a device that converts mechanical energy into electrical energy.

Chemical energy into Heat energy

Burning of fossil fuels such as coal, petrol, diesel, LPG, fireworks, etc. converts chemical energy into heat energy.



Bursting of
firecrackers

Fig. 3.18 Chemical energy $\rightarrow$ Heat,
light and sound energy

xii. Chemical energy into Mechanical energy

In automobiles, the chemical energy of petrol and diesel changes into mechanical energy. Physical exercise is also this type of conversion.



Fig. 3.19 Chemical energy $\rightarrow$ Mechanical energy

xiii. Light energy into Chemical energy

Photosynthesis, exposure of photographic film to light, etc. are some examples that convert light energy into chemical energy.

xiv. Sound energy into Electrical energy

Microphone converts sound energy into electrical energy.

xv. Heat energy into Mechanical energy

The steam engine converts heat energy into mechanical energy.

Energy Chain

The sun is the main source of energy on the earth. Most of the different forms of energy we use in our daily lives have their source in the sun. For example, the electricity that is supplied to our houses to light a bulb or a tube light actually has the sun as its primary source. How does it happen? Let us study about this transfer of energy step by step.

Solar energy is directly or indirectly responsible for the sustenance of life on the

earth. Green plants convert solar energy into chemical energy by the process of photosynthesis. This chemical energy is stored in the plants in the form of sugar.

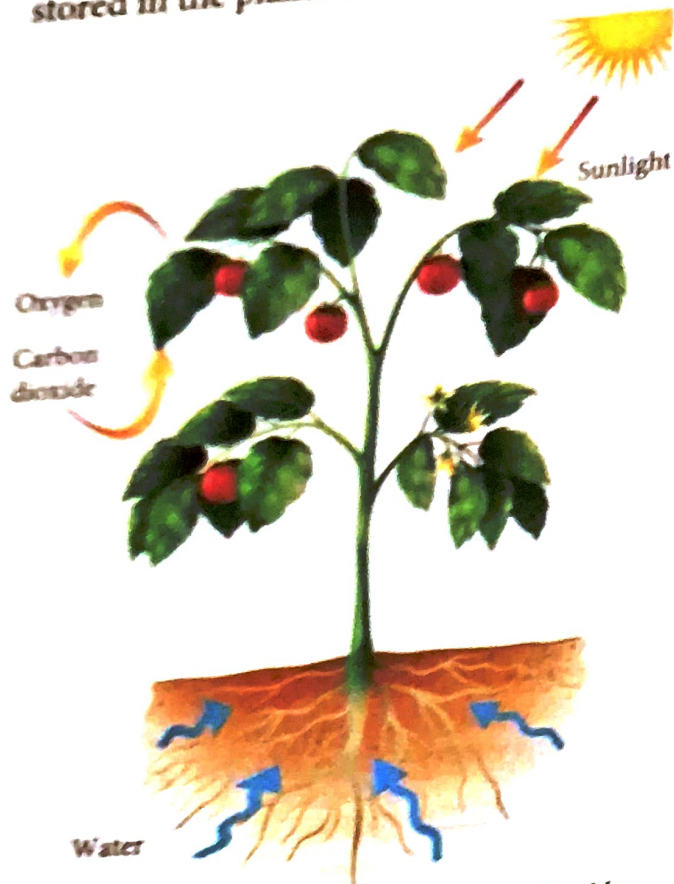


Fig. 3.20 Green plants prepare their own food by photosynthesis

when we eat plants
 4 The chemical energy stored in the plants is transferred to living beings in the form of their food and it remains stored in their body. When dead plants and animals remain buried under the earth for millions of years, 5 (they get converted into fossil fuels like coal) and petroleum. The chemical energy that was stored in the body of plants and animals is now stored in the fossil fuels. 6 When these fuels are burnt in a thermal power plant, electricity is produced, which is supplied to our houses.

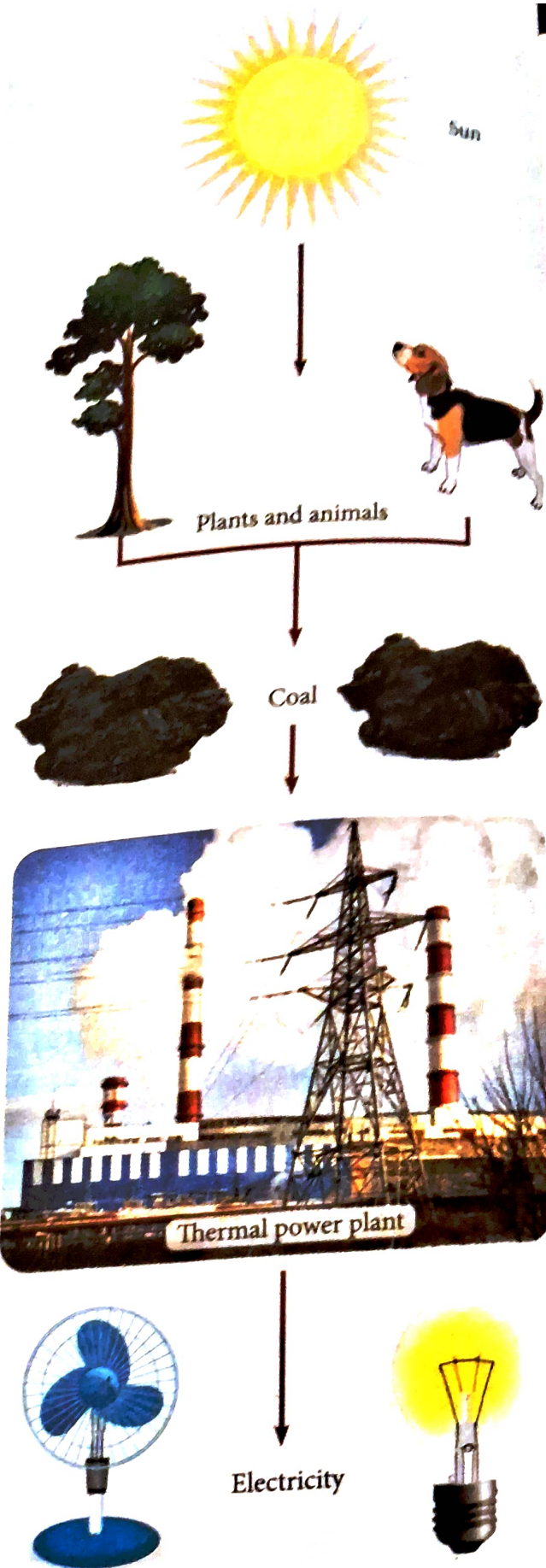
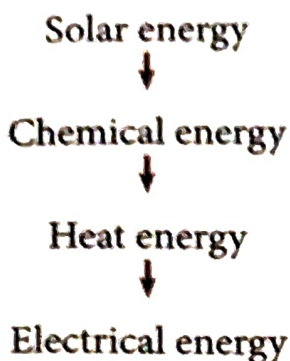


Fig. 3.21 Energy chain

Thus, we can say that the electrical energy that is supplied to our houses actually has its source in the sun. This process of transfer of solar energy into electrical energy makes an energy chain in nature.

In other words, energy chain can be described briefly in the following manner:



Solar energy is also directly converted into electrical energy by solar panels. This electrical energy is used for various purposes like pumping water in the fields to irrigate crops. Solar cooker and solar furnaces directly convert solar energy into heat energy.

Sources of Energy

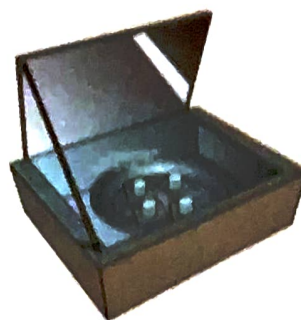
We obtain energy from various sources for different purposes. The various sources of energy can be the sun, flowing water, wind, fossil fuels, biomass, nucleus of an atom, sea waves, etc.

1. Energy from the Sun

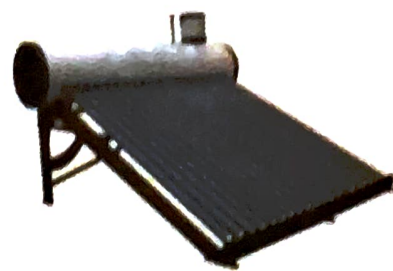
The sun is the ultimate source of energy on the earth.

The energy of the sun makes possible the existence of life on the earth. Plants and animals need solar energy to survive. There are many other applications of solar energy:

Energy



(a) Solar cooker



(b) Solar water heater

Fig. 3.22 Solar appliances using energy from the sun

- Solar cookers and solar water heaters function on getting energy from the sun.
- The solar cells used in artificial satellites, calculators and watches convert stored solar energy into electrical energy.
- Solar energy causes water cycle in nature.
- Solar energy is used to dry clothes and other materials.
- Solar furnaces are used in winters to warm our houses.

EXTRA FACT

One hour of direct sunlight is enough to generate a year's energy for the entire Earth.

2. Energy from Flowing Water

Flowing water is an important source of energy. Big dams constructed over rivers store water. When this stored water is allowed to fall on a turbine, it provides kinetic energy to its blades. The turbine is connected to a generator that converts the kinetic energy into electrical energy. This electrical energy is used for various purposes. Electricity thus produced by water is called hydroelectricity.

3. Energy from Wind

Strong winds possess kinetic energy due to their motion. This energy can turn

the blades of a windmill which is used to pump water from the ground. Nowadays, an electric generator is attached to the axle of rotating vanes that changes its kinetic energy into electrical energy. Sailboats also use the kinetic energy of the wind.



Fig. 3.23 A sailboat using wind energy

4. Energy from Biomass

Garbage, industrial waste, crop residue, sewage, cow dung and other waste materials form biomass. Biomass is decomposed in the absence of oxygen to produce methane gas, which can be used as a fuel. The residue can be used as manure for plants. Biogas is a cheaper and cleaner fuel. It is free from smoke and does not cause air pollution. Gobar gas plants are becoming popular in our villages. Here chemical energy stored in biomass is converted to other forms of energy.



Fig. 3.24 A biogas plant

5. Energy from Fuels

Substances like wood, coal, petrol, diesel or kerosene when burnt produce a large amount of heat. They are used as a fuel. A fuel has chemical energy stored in it. When we burn fuel, its chemical energy is converted into heat and light energy. Coal and petroleum are fossil fuels.

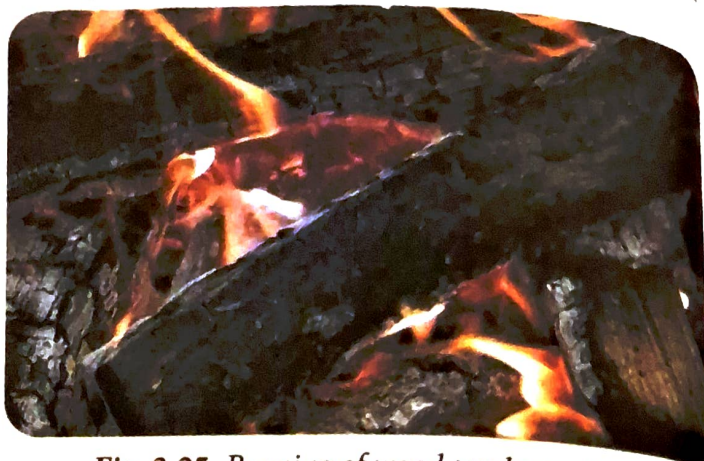


Fig. 3.25 Burning of wood producing large amount of heat

6. Energy from Sea Waves

Scientists have developed many methods to utilise the energy of the sea waves (sea tides). This energy can be used in a turbine to produce electrical energy.

7. Energy from the Nucleus of an Atom

Nuclear energy is obtained by splitting the nucleus of an element like uranium. This process is called **nuclear fission**. It produces a lot of heat energy. This energy is used to produce electricity in nuclear power stations.

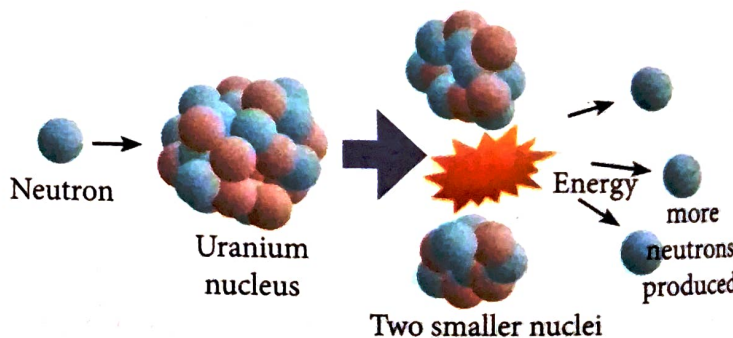


Fig. 3.26 Nuclear fission

EXTRA FACT

Nuclear reactions in the sun are the source of heat and light on the earth. Deep inside a star, hydrogen atoms are constantly converted into helium atoms by a type of nuclear reaction, called fusion.

Work-Energy Relationship

For an object to work, there must be a force acting on it and the object should move in the direction of the force. Why does an object move when a force is applied on it? It is because the force increases the level of energy of the object. It gives the object the ability or capacity to move in the direction of force. When we push a ball lying on the ground, the ball moves forward. In fact, we do work on the ball and the ball acquires kinetic energy.

Similarly, when we compress a spring, we exert a force to do work on it. The spring moves in the direction of the force. As a result, the coils of the spring get compressed and their potential energy increases.

So, we can say that the work done on a system gets stored in it in the form of its energy. That means, Work (W) = Energy (E).

Work done by an object is calculated as the product of the force applied on it and the distance it moves in the direction of the force. It can be expressed as

$$\text{Work (W)} = \text{Force (F)} \times \text{Displacement (s)}$$

$$\text{or } W = Fs$$

$$\text{or } E = W = Fs$$

The amount of work done is actually its stored-up energy that enables it to bounce back, as in the case of a spring, or an apple falling down from a tree.

Conservation of Sources of Energy

Fossil fuels take a few million years to form naturally. There is a limited quantity of such fuels found in nature. Once used, they cannot be renewed. Hence they are called **non-renewable sources of energy**. However, sources of energy like the Sun, water, air, biomass, etc. are found in abundance in nature. They are called **renewable sources of energy**. They are also termed as **alternative sources of energy**. We are, therefore, encouraged to use renewable sources of energy in place of non-renewable energy sources. The judicious use of renewable energy sources is a step taken in the direction of conservation of sources of energy.

Solved Numericals

Example 1: Calculate the potential energy gained by a man weighing 50 kg in climbing up a ladder 15 m high. ($g = 9.8 \text{ m/s}^2$)

Solution:

$$PE = m \times g \times h$$

$$\text{or } PE = 50 \text{ kg} \times 9.8 \text{ ms}^{-2} \times 15 \text{ m} = 490 \times 15 = 7350 \text{ J}$$

Energy

Example 2: A bullet of mass 20 g is moving with a velocity of 600 ms^{-1} . Find the kinetic energy of the bullet.

Solution:

Mass (m) of the bullet = 20 g = 0.020 kg

Velocity (v) of the bullet = 600 ms^{-1}

$$\begin{aligned}\text{Kinetic energy} &= \frac{1}{2}mv^2 \\ &= \frac{1}{2} \times 0.020 \times (600)^2 = 3600 \text{ J}\end{aligned}$$

Example 3: A hammer of mass 2 kg is raised to a height of 3 m. What will be its potential energy?

Solution:

Here mass (m) = 2 kg

Height (h) = 3 m

and g = 9.8 ms^{-2}

$\therefore$ Potential energy = mgh

or PE = $2 \times 9.8 \times 3 = 58.8 \text{ J}$

Let Us Summarise

1. Energy of a body is its capacity to do work. The SI unit of energy is joule (J).
2. Energy appears in many forms such as mechanical energy, heat energy, light energy, electrical energy, magnetic energy, chemical energy, sound energy, nuclear energy and solar energy.
3. Mechanical energy is divided into two parts—kinetic energy and potential energy.
4. The energy possessed by a body by virtue of its motion is called its kinetic energy.
5. The energy possessed by a body by virtue of its position and configuration is called its potential energy.
6. Energy can neither be created nor destroyed. The total energy of the universe remains constant. Energy can be transformed from one form to another.
7. The sun is the ultimate source of energy.
8. The work done on a system gets stored in it in the form of energy. That means, Work (W) = Energy (E).

REVIEW QUESTIONS

Objective Evaluation

A. Choose the correct answer:

- Two bodies A and B of equal mass have their velocities in the ratio 1 : 2. What will be the ratio of their kinetic energies?
a. 2 : 1 b. 1 : 2 c. 1 : 4 d. 4 : 1
- A device that converts light energy into electricity is—
a. electric heater b. electric bulb c. thermocouple d. photocell
- The work done by an object does not depend on the—
a. applied force b. displacement of the body
c. initial velocity of the body d. All of these
- Food that we take in converts—
a. potential energy into kinetic energy b. electrical energy into heat energy
c. chemical energy into heat energy d. chemical energy into muscular energy
- Energy chain consists of—
a. solar energy b. chemical energy c. heat energy d. All of these
- The SI unit of energy is—
a. newton b. joule c. pascal d. second
- Which of the following is not a renewable source of energy?
a. Sun b. Biomass c. Wind d. Petroleum
- The device that converts chemical energy into electrical energy is—
a. dry cell b. electric heater c. electric bulb d. thermocouple
- Potential energy of a person is minimum when—
a. he is standing b. he is sitting on a chair
c. he is sitting on the ground d. he is lying on the ground

B. Fill in the blanks:

- The SI unit of energy is _____.
- The kinetic energy of a body is by virtue of its _____.
- Water cycle is caused due to _____ energy.

4. Strong wind can turn the blades of a windmill because it possesses _____ energy.
5. _____ energy is obtained by splitting the nucleus of an element like uranium.
6. Wood has _____ energy stored in it.
7. An electric bulb converts electric energy into _____ energy.
8. _____ is the ultimate source of energy.
9. If the mass of a body is doubled, its potential energy will be _____.
10. A microphone converts _____ energy into _____ energy.

C. Write True (T) or False (F) against the following statements:

1. A running cat has potential energy.
2. An electric heater converts electrical energy into heat and light energy.
3. A body cannot have more than one type of energy at a time.
4. Solar cells used in calculators convert stored solar energy into electrical energy.
5. Chemical energy is the cause of water cycle.
6. A loudspeaker converts sound energy into electrical energy.
7. A firecracker converts chemical energy into sound, light and heat energy.
8. A falling stone converts kinetic energy into potential energy.

D. With proper reasoning, choose the odd one out:

1. lifting a load, sitting on a chair, compressing a spring, stretching a rubber sheet
2. diesel, petrol, kerosene, photocell
3. room heater, geyser, toaster, thermocouple
4. loudspeaker, microphone, electric bell, stereo system

E. Match the following:

Column A

1. Photosynthesis
2. Dry cell
3. Electromagnet
4. Doorbell
5. Electric heater
6. Table fan
7. Photocell
8. Tubes and bulbs

Column B

- a. Electrical energy → magnetic energy
- b. Electrical energy → sound energy
- c. Solar energy → chemical energy
- d. Chemical energy → electrical energy
- e. Electrical energy → heat energy
- f. Electrical energy → light energy
- g. Electrical energy → mechanical energy
- h. Light energy → electrical energy

R. Name the energy changes in the following examples:

1. Photosynthesis
4. Firecracker
7. Dry cell
10. Photocell
13. Burning candle
16. Cooking gas
19. TV

2. Physical exercise
5. Room heater
8. Thermocouple
11. Table fan
14. Washing machine
17. Generator
20. Electric motor

3. Loudspeaker
6. Microphone
9. Stereo system
12. Tubes and bulbs
15. Electromagnet
18. Windmill

Subjective Evaluation

G. Answer the following questions:

1. How do you define energy?
2. What do you understand by the terms kinetic energy and potential energy?
3. How does kinetic energy of a body change if—
 - i. its mass is doubled?
 - ii. its velocity is doubled?
4. What are the factors on which potential energy depends?
5. State the principle of conservation of energy.

H. Give reasons for the following:

1. Fossil fuels are non-renewable sources of energy.
2. A light and a heavy body can have equal amount of kinetic energy.
3. Light is a form of energy.
4. The Sun is the ultimate source of energy.

Numericals

1. A bullet of mass 100 g is moving with a speed of 200 ms^{-1} . What is the kinetic energy of the bullet?
[Ans: 2000 J]
2. How fast does a man, weighing 60 kg, run so that his kinetic energy is 750 J?
[Ans: 5 m/s]

Life Skills

With the help of your parents and from the electricity bill of your house, find out the total electrical units consumed for the month. What steps will you take to reduce the electrical energy consumption?